PORTFOLIO · UNREAL ENGINE 5 · C++ / BLUEPRINTS

HASSAM ASLAM GAMEPLAY & AI

Gameplay programmer building combat, the Gameplay Ability System, enemy and NPC AI, and multiplayer in UE 5.8. Shipped 12 Fortnite islands as Zeritress.

Field Correspondent	LLM NPC · BT · NavMesh
Ashford Hollow	GAS · enemy AI · netcode
Moshpit Studios	crowd NPC AI · tech anim
Third-person action RPG	combat · progression
GASShooter 5.8 port · UEFN	engine upgrade · shipped

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SOLO PROOF OF CONCEPT · 2026

UE 5.8 | C++ | LLM tool calls | Retrieval with citations | AI Perception

FIELD CORRESPONDENT

A reporter NPC in a newspaper building that you talk to. A language model answers from a handbook it searches and cites; the NPC proposes actions that C++ checks before a behavior tree carries them out; it notices events, remembers them, and files reports that cite what it saw. It runs offline with a scripted model, or with Claude or any OpenAI-compatible server.

THE MODEL ASKS, C++ DECIDES

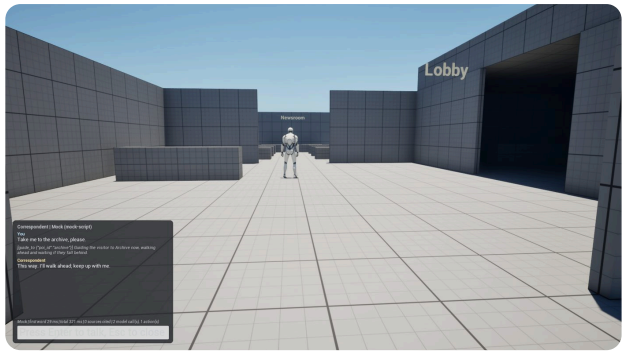
- Five tools: `guide_to`, `follow_player`, `wait_here`, `investigate`, `file_dispatch`. The model never moves the pawn.
- Every call is checked against a whitelist and a NavMesh path test; refusals go back to the model with a reason it explains to the player.
- A new message aborts the running branch: the NPC stops mid-walk in about 0.2 s.

BEHAVIOR AND MEMORY

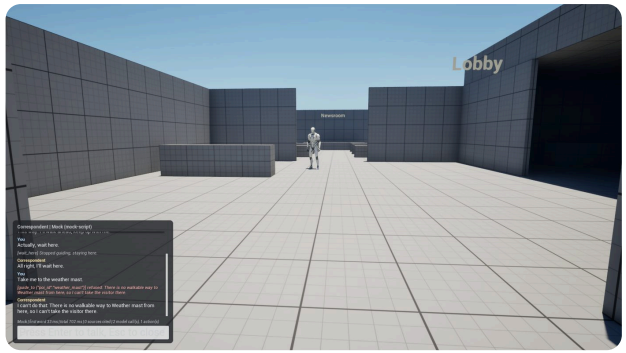
- One behavior-tree branch per mode (Investigate, Guide, Follow, Idle) with six custom C++ nodes; Detour crowd following.
- AI Perception (sight and hearing) turns events into timed memories; a dispatch is refused unless it cites one.

MEASURED

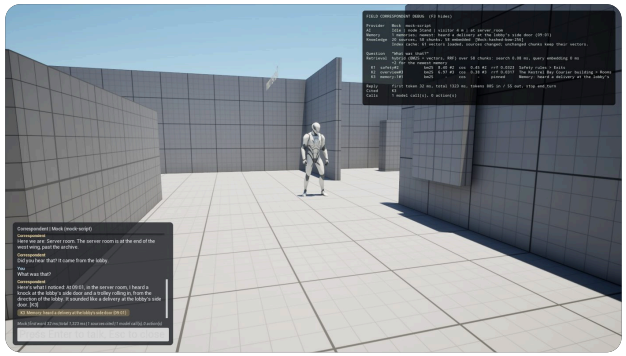
- Hybrid search (BM25 + vectors + rank fusion): 0.044 ms median over 57 chunks.
- 120 automation tests, 50 end-to-end checks passing in a packaged build.



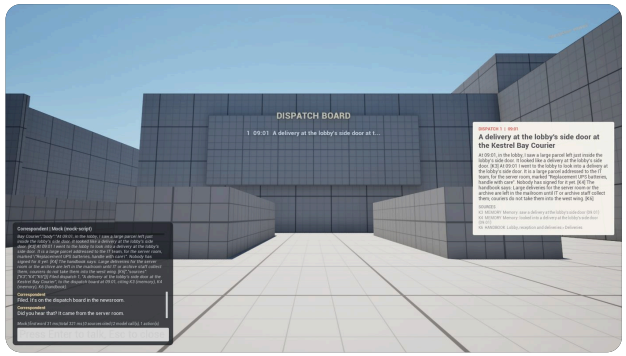
"Take me to the archive": a validated `guide_to` call; it walks ahead and waits if you fall behind.



Interrupted mid-walk, then a refused request: no walkable path to the weather mast.



Debug overlay: retrieved chunks and scores, and the answer citing a heard delivery.



A filed dispatch on the newsroom board, citing memories and the handbook.

SOLO PROJECT · 2026

UE 5.8 | ~18,600 lines C++ | Gameplay Ability System | 2-4 players, server-authoritative

ASHFORD HOLLOW

A medieval top-down pixel-art RPG slice for two to four players over a listen or dedicated server. The server decides every outcome; clients predict only what makes the game feel instant.

COMBAT ON GAS

- Ability system on the PlayerState, custom attribute set, damage and cooldown effects, gameplay tags and hit cues with predicted feedback.
- 3-hit melee combo with input buffering, ranged attack with a predicted local projectile, boss slam and dash with telegraphs synced to server time.

ENEMY AI

- One AI Controller with AI Perception; a behavior tree per enemy type sharing one Blackboard; five custom C++ nodes.
- Wolf pack with flank slots and attack tokens; an archer that picks firing spots with EQS; NavMesh with Detour crowd avoidance.

LAG COMPENSATION, MEASURED

ROUND TRIP	REWIND	HITS SHOWN	GHOST HITS
200 ms	off	61	7 (11%)
200 ms	on	58	**0**
300 ms	off	60	18 (30%)
300 ms	on	58	**1 (2%)**



Day, rain, dusk and night: the server replicates the rule (clock anchor, weather change time), not the value. 0 B/s extra bandwidth.



The Watchtower Guardian's slam telegraph.

```
BT_Wolf
Selector Wolf  [Track Target]
 └ Hunt › Bite or Circle
    ├ Bite [Cooldown 1.4s][Token ×1]
    │  ├ Lunge [Time Limit 1.5s]
    │  ├ Use Ability: melee (GAS)
    │  └ Wait 0.4s
    ├ Circle Target (flank slot)
    └ Wait 0.2s
 └ Return Home › Move To › Idle
```

STUDIO WORK AND FINAL YEAR PROJECT

UE5 · AI Controllers · Behavior trees · Technical animation · Combat

MOSHPIT · ACTION RPG

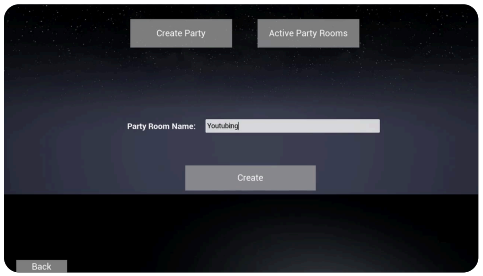
MOSHPIT VIRTUAL CONCERT · MOSHPIT STUDIOS

A multiplayer virtual concert: create or join a party room, then move through a stadium crowd of NPCs while video plays on the stage screens.

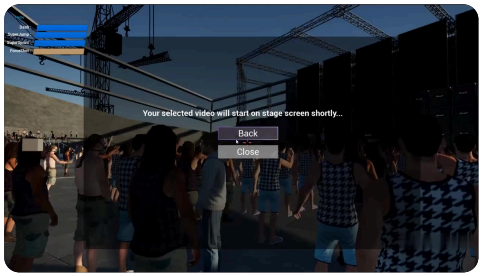
- **My part:** NPC behavior with AI Controllers and behavior trees, technical animation for crowd and player characters, and testing NPCs against player powers.
- **In the build:** Dash, Super Jump, Super Sprint and Force Shot (knocks crowd NPCs into ragdoll); party rooms; an in-game browser that puts a chosen video on the stage.



Inside the crowd.



Creating a party room.



Sending a video to the stage.

THIRD-PERSON ACTION RPG · FINAL YEAR PROJECT

An action game across two levels, a bamboo shrine and a neon city.

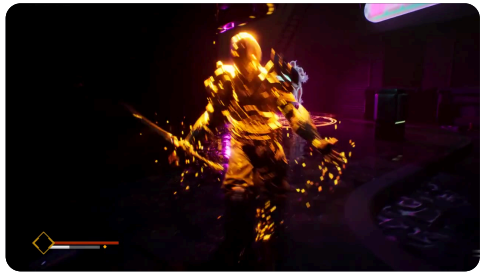
- Melee combat with ice, fire and energy-slash abilities, each with its own screen effect; enemy AI that tracks the player and attacks in groups.
- XP, levelling and skill points; a skill tree in three branches (Combat, Mobility, Attribute) with locked tiers and unlockable moves.



Energy slash in the bamboo shrine.



The three-branch skill tree.



Fire form in the neon city.

Gameplay reels: hassam28.github.io/unreal

Hassam Aslam (Zeritress)

ENGINE WORK AND SHIPPED TITLES

UE 5.3 → 5.8 | GAS weapons | Verse | Live-ops

GASSHOOTER · UEFN

GASSHOOTER PORTED TO UE 5.8

tranek's open-source GAS shooter sample (rifle, shotgun, rocket launcher on GAS) brought from 5.3 to 5.8, with the editor target building cleanly.

- Montage replication moved off the removed `RepAnimMontageInfo.AnimMontage` field to `.Animation` / `GetAnimMontage()`.
- Deprecated NonInstanced abilities moved to InstancedPerActor; ability tags to `Get/SetAssetTags()`; input-pressed prediction keys read from the ability instance.
- Fixed an inverted `ensure` in `GSAT_WaitTargetDataUsingActor` that always returned before registering server-side target-data callbacks.
- Documented how the weapons use GAS: abilities granted on pickup and gated by the current weapon, clip ammo replication paused while firing, predicted weapon switching, one batched RPC per shot.

FORTNITE ISLANDS IN UEFN · AS ZERITRESS, 2023 – PRESENT

12	855M+	44,962	482
islands shipped	minutes played	all-time player peak	published versions

- Six creator partners; flagship PlayStation vs Xbox at 552M minutes; Fort Lock: Soccer and Fashion Show: 50 Players.
- Verse systems: persistent economies and save state, shops and item granting, per-player state tested with 50 players, prestige and ranked progression, NPC customer AI.
- Live-ops at about two updates a week on flagship islands. Full roster at hassam28.github.io.

LET'S BUILD IT

Gameplay, combat and AI programming in Unreal Engine 5. Available remotely, with overlap with US and EU hours.

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